

$\mathbb{A}^{12}$, coordinates $z_0, \dots, z_{11}$

$X = V(e_1, e_3, e_5)$: cone, dim 9, complete intersection, normal
smooth where $\#\{z_j\} \geq 3$ (Vandermonde rows $1, z_j^2, z_j^4$ of rank 3)

Lucas: $|A|$ even $\Rightarrow e_1 = e_3 = e_5 \equiv 0$ on P_A (each term has a factor $\binom{\text{even}}{\text{odd}}$)

$|A|$ odd $\Rightarrow e_1 = u + v$, so $P_A \cap X = L$; $\text{Sing}(X) = \bigcup P_A$ over even $|A|$: dim 2, codim 7
 $1023 = 66 + 495 + 462$ planes (A and its complement give the same plane)



$|A| = 2$: $\binom{12}{2} = 66$ planes

$|A| = 4$: $\binom{12}{4} = 495$ planes

$|A| = 6$: $\frac{1}{2} \binom{12}{6} = 462$ planes

thick line: the diagonal $L = \{(u, \dots, u)\}$, contained in every $P_A = \{z_j = u \ (j \in A), \ z_j = v \ (j \notin A)\}$